Introduction to Cryptography

Lecture 13–14: Modern Cryptography and Symmetric Encryption Principles

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From Classical to Modern Cryptography

Classical cryptographic techniques studied so far include:

- Caesar cipher
- Monoalphabetic substitution
- Playfair cipher
- Vigenère cipher
- Hill cipher
- Transposition ciphers
- Rotor machines

These techniques provide the foundation for modern cryptography.
However, modern cryptography is designed for:

- Digital information
- Computer-based processing
- Large key spaces
- Automated cryptanalysis resistance
- High-speed implementation

Classical vs Modern Cryptography

Aspect	**Classical**	**Modern**
Data	Letters / characters	Binary data
Key Space	Generally small	Very large
Operations	Simple substitution / transposition	Complex combinations of transformations
Processing	Often character-oriented	Bit / byte / block-oriented
Cryptanalysis	Frequency analysis often effective	Designed against systematic attacks
Implementation	Manual or simple mechanical	Software / hardware based

Modern Cryptography

Modern cryptographic algorithms are generally based on:

1. Mathematical transformations
2. Large key spaces
3. Repeated rounds of processing
4. Substitution and permutation
5. Key-dependent transformations
6. Efficient implementation

The objective is to produce a ciphertext that reveals as little useful information as possible about the plaintext and the key.

Three Dimensions of Cryptographic Systems

Cryptographic systems can be classified according to three dimensions:

1. **Type of operations**
   - Substitution
   - Transposition
2. **Number of keys**
   - Symmetric / single-key
   - Asymmetric / two-key
3. **Processing of plaintext**
   - Block cipher
   - Stream cipher

Symmetric Encryption

Symmetric encryption is also known as:

- Conventional encryption
- Secret-key encryption
- Single-key encryption

The same secret key is shared between the communicating parties.



$$C = E_K(P)$$

$$P = D_K(C)$$

Symmetric Cipher Model

A symmetric encryption system has five basic ingredients:

1. Plaintext
2. Encryption algorithm
3. Secret key
4. Ciphertext
5. Decryption algorithm

$$P \xrightarrow{E_K} C \xrightarrow{D_K} P$$

where:

$$P = \text{Plaintext}$$

$$C = \text{Ciphertext}$$

The Five Basic Ingredients

Component	Meaning
Plaintext	Original intelligible message or data.
Encryption Algorithm	Performs transformations on plaintext.
Secret Key	Controls the operation of the encryption algorithm.
Ciphertext	Encrypted and unintelligible form of the message.
Decryption Algorithm	Recovers plaintext from ciphertext using the key.

Encryption and Decryption

Encryption:

$$C = E_K(P)$$

Decryption:

$$P = D_K(C)$$

For a valid cryptosystem:

$$D_K(E_K(P)) = P$$

The encryption algorithm and decryption algorithm are designed so that the legitimate receiver can recover the original message.

Plaintext $\rightarrow$ Encryption $\rightarrow$ Ciphertext $\rightarrow$ Decryption $\rightarrow$ Plaintext

The Role of the Key

The key controls the transformation performed by the algorithm.
For the same plaintext:

$$E_{K_1}(P) \neq E_{K_2}(P)$$

in a well-designed cryptographic system.

Therefore:

- The algorithm may be publicly known.
- The ciphertext may be observed by an attacker.
- The plaintext may sometimes be partially known.
- The secret key must remain protected.

Key secrecy is fundamental to symmetric encryption

Key Space

If the key has n bits, the total number of possible keys is:

$$\boxed{2^n}$$

Examples:

Key Length	Number of Possible Keys
8 bits	$2^8 = 256$
16 bits	$2^{16} = 65,536$
32 bits	2^{32}
56 bits	2^{56}
128 bits	2^{128}

A larger key space generally increases resistance to exhaustive key-search attacks.

Brute-Force Key Search

Suppose an attacker does not know the key.
The attacker can attempt:

$$K_1, K_2, K_3, \ldots, K_{2^n}$$

until a valid key is found.
For an ideal key space, the average number of trials is approximately:

$$\boxed{2^{n-1}}$$

Therefore, increasing the key length dramatically increases the search space.

Kerckhoffs' Principle

A fundamental principle of cryptographic system design is:

The security of a cryptosystem should depend on the secrecy of the key, not on the secrecy of the algorithm.

Thus an attacker may know:

- The encryption algorithm
- The decryption algorithm
- The ciphertext
- The implementation details

but should not know the secret key.

Substitution and Transposition

Substitution

- Replaces one symbol with another.
- Example:

$$A \to D, \quad B \to E$$

Transposition

- Rearranges the positions of symbols.
- Symbols themselves are not replaced.

Modern ciphers generally combine these types of transformations in more complex structures.

Product Cipher

A product cipher combines multiple transformations.
For example:

$$\text{Substitution} \rightarrow \text{Permutation} \rightarrow \text{Substitution} \rightarrow \text{Permutation}$$

Mathematically:

$$C = T_n(T_{n-1}(\cdots T_2(T_1(P))\cdots))$$

where each T_i represents a transformation.

The combination of transformations can produce a much stronger cipher than an individual transformation.

Confusion

The concept of **confusion** is associated with making the relationship between the key and ciphertext as complex as possible.

A cipher should make it difficult for an attacker to determine:

$$\text{Key} \longrightarrow \text{Ciphertext relationship}$$

Substitution operations are commonly associated with providing confusion.

Confusion hides the relationship involving the key

Diffusion

Diffusion refers to spreading the influence of plaintext statistics over many ciphertext elements.

A desirable property is that a small change in plaintext should affect many ciphertext bits.

Example concept:

$$P \rightarrow \boxed{\text{Cipher}} \rightarrow C$$

Changing one input bit should ideally produce significant changes in the resulting ciphertext.

Diffusion spreads the influence of plaintext

Confusion vs Diffusion

Confusion	**Diffusion**
Complicates the relationship between key and ciphertext	Spreads the influence of plaintext statistics
Associated with substitution	Associated strongly with permutation/transposition
Makes key relationships difficult to analyze	Makes plaintext statistical patterns difficult to identify

Strong modern ciphers aim to provide both.

Block Cipher

A block cipher processes plaintext in fixed-size blocks.
If the block size is n bits:

$$P_i \in \{0,1\}^n$$

and:

$$C_i = E_K(P_i)$$

Decryption gives:

$$P_i = D_K(C_i)$$

Examples of important block ciphers include:

- DES
- Triple DES
- AES

Stream Cipher

A stream cipher processes plaintext as a continuous stream.
Conceptually:

$$P_1, P_2, P_3, \ldots$$

are combined with a generated key stream:

$$K_1, K_2, K_3, \ldots$$

For a simple XOR-based model:

$$C_i = P_i \oplus K_i$$

and:

$$P_i = C_i \oplus K_i$$

because:

Block Cipher vs Stream Cipher

	Block Cipher	**Stream Cipher**
Processing	Fixed-size blocks	Continuous stream
Unit	Block of bits	Usually bits or bytes
Typical Structure	Multiple rounds	Key-stream generation
Examples	DES, AES	RC4 (historical)

Modern Block Cipher Structure

A modern block cipher generally applies a sequence of transformations.

$$P \to R_1 \to R_2 \to R_3 \to \cdots \to R_N \to C$$

Each round may use a different round key:

$$K_1, K_2, \ldots, K_N$$

The round keys are normally derived from the main secret key through a key-scheduling process.

Why Multiple Rounds?

A single transformation may not provide sufficient complexity.
Multiple rounds allow:

- Repeated mixing of data.
- Repeated use of key-dependent transformations.
- Greater diffusion.
- Greater confusion.
- Increased resistance to cryptanalysis.

Conceptually:

$$P \xrightarrow{K_1} R_1 \xrightarrow{K_2} R_2 \xrightarrow{K_3} \cdots \xrightarrow{K_N} C$$

Feistel Structure

The Feistel structure is an important design approach for symmetric block ciphers.
The input block is divided into two halves:

$$P = L_0 \parallel R_0$$

For a typical round:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

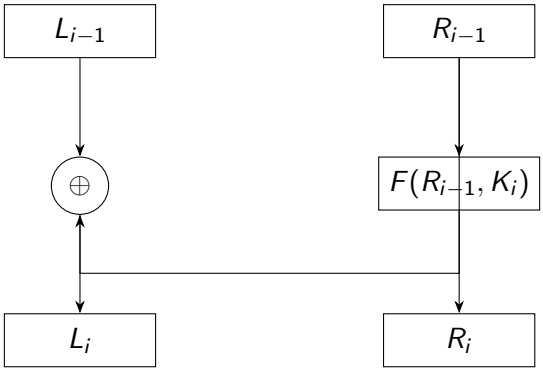
where:

$$F = \text{round function}$$

and:

$$K_i = \text{round key}$$

One Feistel Round



$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Why is Feistel Structure Important?

Important characteristics include:

- The block is divided into two halves.
- A round function operates on one half.
- The result is combined with the other half using XOR.
- The halves exchange roles between rounds.
- Multiple rounds provide increasing complexity.

A major historical example is:

DES

which uses a Feistel structure.

Modern Symmetric Algorithms

Important symmetric algorithms include:

- DES
- Triple DES
- AES

DES

- Historical importance
- Feistel-based design
- 64-bit block size
- 56-bit effective key

AES

- Modern standard
- 128-bit block size
- 128-, 192-, or 256-bit keys
- Not based on a Feistel structure

Classroom Numerical

Question 1

An encryption algorithm uses a 16-bit key.

1. Calculate the total number of possible keys.
2. Calculate the average number of trials required by an exhaustive key search.

Solution

$$2^{16} = 65,536$$

Average trials:

$$\frac{2^{16}}{2} = 2^{15} = 32,768$$

Think About It

Question
Suppose an attacker knows:

- The encryption algorithm
- The ciphertext
- The implementation
- Several plaintext-ciphertext pairs

but does not know the secret key.

Question:
Should a well-designed symmetric cryptosystem still remain secure?

Yes — security should primarily depend on the secrecy of the key.

Feistel Cipher Structure

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

From Feistel Structure to DES

References

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2. William Stallings, *Computer Security: Principles and Practice*, Pearson.
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4. Alfred J. Menezes, Paul C. van Oorschot, and Scott A. Vanstone, *Handbook of Applied Cryptography*, CRC Press.